

# AI-Powered YouTube Shorts Automation System

A fully automated, scheduled content system that researches a topic, writes a script, produces a professional voiceover, sources matching stock footage, assembles a finished vertical video with background music, and publishes it — from a single daily trigger, with no manual editing.

|              |                                                                                                                                        |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------|
| What it does | Automates script writing, voice synthesis, stock-footage sourcing, video assembly, rendering, publishing, and content-history tracking |
| Tools used   | n8n · AI Agent (LLM) · ElevenLabs · Pexels · Cloudinary · Shotstack · Google Sheets · YouTube Data API                                 |
| Who benefits | Content Creators, Marketing & Growth Teams, and Agencies managing multiple channels                                                    |
| Key features | AI script generation · automated voiceover · smart clip sequencing · dynamic timing · one-click multi-stage publishing                 |
| Built by     | Workflow Automation & Systems Integration — Portfolio Project                                                                          |

## What Is This Project?

Producing short-form video content consistently is slow when every step is manual: researching a topic, writing a script, recording a voiceover, finding footage, editing, and publishing — repeated for every single video. Most pipelines break down somewhere in that chain.

This system was built to remove every manual step. A daily trigger kicks off the process: an AI agent writes a script and a set of visual search prompts, a voice synthesis API produces the narration, matching stock footage is sourced automatically, and a cloud rendering service assembles everything — title card, sequenced clips, narration, and background music — into a finished vertical video that publishes itself. **From topic to published video, no manual editing is required.**

*Built for continuous operation: the pipeline tracks every topic it has already covered and reads that history back into the AI agent on every run, so scheduled, unattended operation never repeats itself.*

9

### Pipeline Stages

Script to publish, fully chained

0

### Manual Edits

Every video assembled automatically

AI

### Script Writer

Agent generates title, script & prompts

24/7

### Scheduled

Runs unattended on a daily trigger

## The Pipeline at a Glance

The system runs in two connected phases. Phase 1 covers research, writing, and voice production. Phase 2 covers footage sourcing, assembly, rendering, and publishing. Together they cover the full content lifecycle:

Phase 1 begins the moment the daily schedule trigger fires. Everything that follows — checking topic history, writing the script, and producing the narration — happens automatically.

1

**Daily trigger reads content history**

A scheduled trigger fires once a day. The system reads the last 30 published topics from a tracking spreadsheet so the AI never repeats a subject.

2

**An AI agent writes the script**

An LLM-powered agent, paired with a structured output parser, generates a viral-style title, a ~150-word voiceover script, and a set of stock-footage search prompts — returned as validated JSON, not free-form text.

3

**The script is converted into narration**

The generated script is sent to a text-to-speech API, which returns a natural-sounding voiceover as an audio file, ready for production.

4

**The narration is hosted for the renderer**

The synthesized audio is uploaded to a cloud media host to obtain a publicly accessible URL that the downstream rendering service can consume directly.

Phase 2 picks up the moment the narration is ready — sourcing footage, assembling the edit, rendering in the cloud, and publishing the result without anyone touching a file.

5

**Matching stock footage is sourced**

The AI-generated visual prompts are used to search a stock video library for vertical-orientation b-roll clips that match each beat of the script.

6

**The video is assembled automatically**

A JSON-defined timeline sequences a title card, b-roll clips, narration, and background music into a 1080×1920 vertical video. Clip timing adapts to each source clip's real duration and to the narration's actual length, so playback stays continuous and audio stays in sync start to finish.

7

**The edit is rendered in the cloud**

The completed timeline is submitted to a cloud rendering service. The workflow polls the render job on an interval and only proceeds once processing is confirmed complete.

8

**The finished video is published**

The rendered video is automatically uploaded and published to YouTube — no manual upload, no manual metadata entry.

9

**The result is logged for next time**

The published title, timestamp, and video URL are appended back to the tracking spreadsheet, closing the loop for the next run's history check.

## The Tools Working Together

Eight tools work together across both phases. Each one handles a specific part of the content lifecycle:

|          |                                                                                                                                                                                                                                           |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>n</b> | <b>n8n — The automation engine</b><br>n8n is the workflow platform that powers both phases. It connects every tool, runs the script and assembly logic, manages render polling, and orchestrates publishing from start to finish.         |
| <b>A</b> | <b>AI Agent — The script writer</b><br>An LLM-powered agent with a structured output parser writes the title, script, and visual search prompts for every video, returning clean, schema-validated JSON the pipeline can act on directly. |
| <b>E</b> | <b>ElevenLabs — The voice synthesis engine</b><br>Converts the generated script into a natural-sounding voiceover, giving every video a consistent, professional narration track.                                                         |
| <b>P</b> | <b>Pexels — The stock footage library</b><br>Searched using the AI's own visual prompts to source vertical b-roll clips that match the narrative beat for beat.                                                                           |
| <b>C</b> | <b>Cloudinary — The media hosting layer</b><br>Hosts the generated narration audio so the rendering service has a public URL to pull from during assembly.                                                                                |
| <b>S</b> | <b>Shotstack — The cloud rendering engine</b><br>Assembles the title card, footage, narration, and background music into a finished vertical video from a single JSON-defined timeline.                                                   |
| <b>G</b> | <b>Google Sheets — The content history layer</b><br>Stores every published topic and result, giving the AI agent the history it needs to avoid repeating a subject on the next run.                                                       |
| <b>Y</b> | <b>YouTube Data API — The publishing layer</b><br>Receives the finished video and handles the upload and publish step automatically once rendering is confirmed complete.                                                                 |

## Who Benefits from This System?

This system was built with the realities of consistent content production in mind. Three groups benefit directly:

### Content Creators

A daily trigger is the entirety of your production responsibility. The script, voiceover, footage, and finished video are produced and published automatically.

### Marketing & Growth Teams

Consistent daily output without hiring a production team. Every video follows the same structure, pacing, and quality bar — automatically.

### Agencies & Multi-Channel Operators

The same pipeline runs in parallel across topics and channels. Growing volume does not mean growing headcount.

## Why This Matters

- ✓ **Zero time spent on manual production**  
The entire script-to-publish process runs from a single scheduled trigger. No writing, recording, editing, or uploading by hand.
- ✓ **Every video is consistent and on-format**  
Because the timeline is generated programmatically, every video matches the same vertical format, pacing, and structure — with no manual editing inconsistencies.
- ✓ **Smart sequencing removes editing guesswork**  
Clip timing adapts automatically to each source clip's real length and the narration's actual duration, so footage cuts continuously and audio never gets cut short.
- ✓ **It scales without adding production overhead**  
Whether the pipeline produces one video a day or runs across multiple topics in parallel, every run follows the same automated process.